Fiscal Year:	FY 2019	**Task Last Updated:**	FY 03/29/2021
PI Name:	Turek, Fred W Ph.D.		
Project Title:	HERO Twin Astronaut Study Consortium (TASC) Project: Metagenomic Sequencing of the Bacteriome in GI Tract of Twin Astronauts		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	60201-2919	**Congressional District:**	9
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2013-14 HERO NNJ13ZSA002N-TWINS
Start Date:	04/14/2014	**End Date:**	12/31/2018
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:	Charles, John	**Contact Phone:**	281-483-7224
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 12/31/2018 per NSSC information (Ed., 5/25/18) NOTE: End date changed to 4/13/2018 per NSSC information (Ed., 6/29/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Vitaterna, Martha Ph.D. (Northwestern University)		
Grant/Contract No.:	NNX14AH26G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	High throughput DNA/RNA-based technology has led to a revolution in our understanding of the complexity and diversity of the microbiota communities that inhabit many parts of the human body, including those in the gastrointestinal tract (GI). Disruption of normal bacterial communities in the intestine (i.e., dysbiosis) has been linked to several chronic inflammatory diseases and pathological conditions. The complex ecological microbiology community that inhabits the human GI tract influences normal physiology and behavior and the susceptibility to disease. Despite the clear importance of the GI microbiota for maintaining overall health and influencing disease state, the effects of the spaceflight environment on the human microbiome remain unknown. Thus, it is imperative that studies be carried out on long-term missions in space so that any adverse changes can be identified and countermeasures can be employed to insure the safety and health of our astronauts on extended spaceflight missions. This NASA Human Exploration Research Opportunities (HERO) project is centered around sequencing of the microbiome of twin astronauts. It is also complementary to other biomolecular research projects which seek to integrate -omics-based studies on the effects of spaceflight and will involve data-sharing and data analyses with the other Twins research teams.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The spaceflight environment is both unusual and stressful in ways that challenge gastrointestinal and immune function. However, adverse effects of stress on gastrointestinal (GI) and immune function are present in life on Earth as well. The microbiome of the GI tract is generally understood to impact both GI and immune function, but not yet to a degree to which predictions can be made of what microbiome changes may occur in response to stress, or what microbiome changes may predict GI or other health outcomes. By examining the microbiome changes in the context of the other twins studies -omics data, we hope to enhance our understanding of the microbiome's role in shaping responses to both the spaceflight environment and other, Earth-bound stresses. Further, this study has highlighted some unanticipated spaceflight-associated changes, such as the lack of loss of diversity in the gut microbiome, that have been replicated in animal and other studies. This Twins Study has served to generate new, novel hypotheses of the impacts of long-duration spaceflight on multiple physiological systems.
Task Progress:	SAMPLES ANALYZED. Stool samples were obtained at nine dates from each twin over the course of nearly 2 years. For the “space twin,” two samples were collected prior to launch, four samples were collected during the nearly one-year mission, and three samples were collected following return to Earth. For the “ground twin,” all samples were collected on Earth at intervals spanning the same time period. Shotgun metagenomic sequencing of DNA isolated from the stool samples was used to identify microorganisms and their genes. A. RESULTS: RELATIVE STABILITY OF THE GUT MICROBIOME. The bacterial "fingerprints" (species composition and relative abundance characteristics of an individual) of both twins were maintained over the two-year study period. The space twin's Shannon index (a measure of diversity) was maintained, and may have increased over time during the spaceflight, while the ground twin's slightly declined. This was unexpected for the enclosed, limited-introduction environment of the International Space Station (ISS). B. RESULTS: SHIFTS IN THE BALANCE OF BACTERIAL PHYLA. Firmicutes and Bacteroidetes represent the most abundant bacterial phyla inhabiting the human gastrointestinal tract, accounting for over 90% of the bacteria present. Both the space twin and the ground twin had a ratio of Firmicutes: Bacteroidetes (F:B) within a normal, healthy range at all times. The ground twin and the space twin had elevated F:B ratios at the second timepoint, pre-flight. However, the space twin’s F:B ratio was more than three times that of the initial ground sample for the duration of the spaceflight, then returned to pre-launch levels following return to Earth. A change in the gut bacterial community structure is associated with spaceflight, but appears to recover when the astronaut returns to Earth. C: RESULTS: DIFFERENTIAL ABUNDANCE DURING FLIGHT PERIOD. The relative abundance of specific bacterial species and of specific bacterial genes were both analyzed during the flight period and the on-ground. Far more differentially abundant taxa and genes were identified in the "space twin" than the "ground twin." This highlights the functional changes in the gut microbiome that occurred in response to the spaceflight environment. Metabolomic analysis identified enrichment in microbially-derived phenyls in samples from the spaceflight period.
Bibliography Type:	Description: (Last Updated: 07/16/2025)
Articles in Other Journals or Periodicals	Danko DC, Singh N, Butler DJ, Mozsary C, Jiang P, Keshavarzian A, Maienschein-Cline M, Chlipala G, Afshinnekoo E, Bezdan D, Garrett-Bakelman F, Green SJ, Turek FW, Vitaterna MH, Venkateswaran K, Mason CE. "Genetic and immunological evidence for microbial transfer between the International Space Station and an astronaut." bioRxiv Preprint server. Preprint manuscript posted December 2, 2020. https://doi.org/10.1101/2020.11.10.376954 , Dec-2020
Articles in Other Journals or Periodicals	Vitaterna MH Jiang P. "The NASA Twins Study: The microbiome in space, from the Apollo missions to present." Physiology News Magazine. 2020 Winter;117:16-7. https://doi.org/10.36866/pn.117.16 , Jan-2020
Articles in Peer-reviewed Journals	Garrett-Bakelman FE, Darshi M, Green SJ, Gur RC, Lin L, Macias BR, McKenna MJ, Meydan C, Mishra T, Nasrini J, Piening BD, Rizzardi LF, Sharma K, Siamwala JH, Taylor L, Vitaterna MH, Afkarian M, Afshinnekoo E, Ahadi S, Ambati A, Arya M, Bezdan D, Callahan CM, Chen S, Choi AMK, Chlipala GE, Contrepois K, Covington M, Crucian BE, De Vivo I, Dinges DF, Ebert DJ, Feinberg JI, Gandara JA, George KA, Goutsias J, Grills GS, Hargens AR, Heer M, Hillary RP, Hoofnagle AN, Hook VYH, Jenkinson G, Jiang P, Keshavarzian A, Laurie SS, Lee-McMullen B, Lumpkins SB, MacKay M, Maienschein-Cline MG, Melnick AM, Moore TM, Nakahira K, Patel HH, Pietrzyk R, Rao V, Saito R, Salins DN, Schilling JM, Sears DD, Sheridan CK, Stenger MB, Tryggvadottir R, Urban AE, Vaisar T, Van Espen B, Zhang J, Ziegler MG, Zwart SR, Charles JB, Kundrot CE, Scott GBI, Bailey SM, Basner M, Feinberg AP, Lee SMC, Mason CE, Mignot E, Rana BK, Smith SM, Snyder MP, Turek FW. "The NASA Twins Study: A multidimensional analysis of a year-long human spaceflight." Science. 2019 Apr 12;364(6436):eaau8650. https://science.sciencemag.org/content/364/6436/eaau8650.long ; PubMed PMID: 30975860 ; PMCID: PMC7580864 , Apr-2019

Articles in Peer-reviewed Journals	Afshinnekoo E, Scott RT, MacKay MJ, Pariset E, Cekanaviciute E, Barker R, Gilroy S, Hassane D, Smith SM, Zwart SR, Nelman-Gonzalez M, Crucian BE, Ponomarev SA, Orlov OI, Shiba D, Muratani M, Yamamoto M, Richards SE, Vaishampayan PA, Meydan C, Foox J, Myrrhe J, Istasse E, Singh N, Venkateswaran K, Keune JA, Ray HE, Basner M, Miller J, Vitaterna MH, Taylor DM, Wallace D, Rubins K, Bailey SM, Grabham P, Costes SV, Mason CE, Beheshti A. "Fundamental biological features of spaceflight: Advancing the field to enable deep-space exploration." Cell. 2020 Nov 25;183(5):1162-84. Review. https://doi.org/10.1016/j.cell.2020.10.050 ; PMID: 33242416 , Nov-2020